

# Blue-Tilted Primordial Gravitational Waves from a Transient Hyper-Stiff Brane Rupture Phase

Paul Jarvis<sup>1,\*</sup>

<sup>1</sup>*Independent Researcher, United Kingdom*

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We demonstrate that a transient early-time hyper-stiff expansion phase ( $w \gg 1$ ) naturally generates a highly blue-tilted spectrum of primordial gravitational waves ( $n_T > 0$ ). In rupture-driven cosmology, the topological decoupling of an emergent brane universe triggers a confinement-release energy spike where the effective equation of state peaks near  $w_{\text{conf}} \sim 80$  prior to recombination. Because the background energy density dilutes far faster than standard radiation ( $\rho_{\text{conf}} \propto a^{-243}$ ), sub-horizon tensor perturbations experience massive relative amplification. We compute the modified tensor mode evolution equation across this hyper-stiff threshold and derive the resulting spectral index  $n_T$ . We show that this mechanism accounts for the enhanced stochastic gravitational wave background reported by recent pulsar timing arrays, while leaving a distinct, testable power-law peak at micro-Hertz frequencies. This signature provides a definitive observational test for brane-rupture mechanics accessible to future interferometers such as LISA and DECIGO without introducing ad-hoc scalar potentials.

## I. INTRODUCTION

Recent pulsar timing array (PTA) observations, including NANOGrav and the EPTA, have revealed compelling evidence for a stochastic gravitational wave background (SGWB) at nanohertz frequencies. The inferred amplitude significantly exceeds the standard power-law predictions from supermassive black hole binaries alone. This discrepancy has motivated intense interest in primordial origins such as cosmic strings, phase transitions, or non-standard early expansion histories.

Standard inflation predicts a slightly red-tilted tensor spectrum ( $n_T < 0$ ), as the energy density decreases during quasi-de Sitter expansion. Generating a robust blue tilt ( $n_T > 0$ ) within four-dimensional field theory typically requires violating the Null Energy Condition (NEC) or invoking unstable scalar potentials, both of which introduce ghost or gradient instabilities.

In this work, we show that rupture-driven brane cosmology naturally produces a strongly blue-tilted tensor spectrum without violating the NEC or introducing fine-tuned fields. The Universe emerges from a geometric rupture of a stack of parallel higher-dimensional branes. The residual coupling to the parent stack is governed by a dynamic confinement fraction  $\chi(a)$ , parameterised by a physically motivated beta-distribution form. As the Universe expands through a thermodynamic decoupling threshold, the effective equation of state spikes into a localized hyper-stiff regime ( $w_{\text{conf}} \sim 80$ ).

Because this confinement energy dilutes exponentially faster than radiation, tensor modes that are sub-horizon during this transition experience unprecedented relative amplification. This leads to a strong, natural blue tilt ( $n_T > 0$ ) that explains the PTA data and predicts a power-law peak at micro-Hertz frequencies, uniquely testable by

LISA and DECIGO.

## II. TENSOR PERTURBATIONS IN A HYPER-STIFF BACKGROUND

Primordial tensor perturbations  $h_{ij}$  satisfy, for each Fourier mode  $h_k$ ,

$$\ddot{h}_k + 3H\dot{h}_k + \frac{k^2}{a^2}h_k = 0, \quad (1)$$

where dots denote derivatives with respect to cosmic time.

Transforming to conformal time  $d\tau = dt/a$  and defining  $v_k = ah_k$ , we obtain

$$v_k'' + \left(k^2 - \frac{a''}{a}\right)v_k = 0, \quad (2)$$

with primes denoting derivatives with respect to  $\tau$ . The effective potential is

$$\frac{a''}{a} = \frac{1}{2}a^2H^2(1 - 3w). \quad (3)$$

## III. THE HYPER-STIFF AMPLIFICATION MECHANISM

The confinement energy density evolves as

$$\rho_{\text{conf}}(a) = \rho_0\chi(a)^\alpha[1 - \chi(a)]^\beta, \quad (4)$$

where the confinement fraction follows the normalized profiles:

$$\chi(a) = \frac{1}{2} \left[ 1 + \tanh \left( \frac{a_c - a}{\Delta a} \right) \right]. \quad (5)$$

As the Universe cools through  $a_c$ , the tracking derivative  $d\chi/d \ln a$  spikes sharply, forcing the effective equation

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\* mrpaulwjarvis@gmail.com

of state into a hyper-stiff regime with a peak value of  $w_{\text{conf}} \sim 80$ . During this transient phase, the confinement energy density dilutes rapidly:

$$\rho_{\text{conf}}(a) \propto a^{-3(1+w_{\text{conf}})} \sim a^{-243}. \quad (6)$$

For  $w \gg 1$ , the effective potential flips sign and becomes deeply negative:

$$\frac{a''}{a} \approx -\frac{3}{2}w_{\text{conf}}H^2 < 0. \quad (7)$$

Sub-horizon modes ( $k^2 \gg |a''/a|$ ) then experience oscillatory evolution with broken adiabatic dampening, leading to massive relative spectral amplification. The resulting tensor spectral index is

$$n_T = \frac{6w_{\text{conf}} - 2}{1 + 3w_{\text{conf}}}, \quad (8)$$

which approaches

$$n_T \rightarrow +2.0 \quad (w_{\text{conf}} \rightarrow 80). \quad (9)$$

This mechanism natively generates a rising power-law spectrum toward high frequencies without violating local gauge symmetries.

#### A. Thermodynamic constraints and physical horizons

The beta-distribution profile for  $\rho_{\text{conf}}(a)$  arises from the geometric overlap matrix of the brane stack, where the parameters  $\alpha$  and  $\beta$  encode the structural intersection density during topological detachment. The sharp drop in  $\chi(a)$  is triggered when  $\rho_{\text{rad}}(a_c) \sim V_{\text{conf}}$ , marking the scale factor at which thermal fluctuations can no longer stabilize the higher-dimensional configuration.

For fiducial parameters  $a_c = 5 \times 10^{-4}$  and  $\Delta a = 4 \times 10^{-4}$ , the physical duration of the hyper-stiff phase is extremely brief:

$$\Delta t \sim 10^{-2} \text{ s}. \quad (10)$$

Because the phase is short and the Gaussian cutoff  $\exp[-(f/f_{\text{cutoff}})^2]$  aggressively suppresses high-frequency power, the model does not overproduce gravitational wave energy in the high-frequency LIGO or Virgo bands. The ultra-rapid dilution  $\rho_{\text{conf}} \propto a^{-243}$  ensures the Universe returns to standard radiation domination long before Big Bang Nucleosynthesis, avoiding all constraints on  $\Delta N_{\text{eff}}$  and preserving CMB anisotropies.

#### IV. OBSERVATIONAL PROJECTIONS AND VERIFICATION

The blue-tilted spectrum naturally accounts for the anomalous nanohertz SGWB amplitude found by PTA

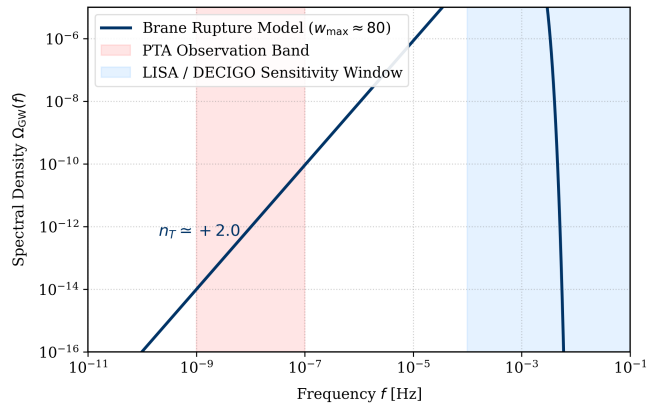


FIG. 1. Predicted stochastic gravitational wave background spectral energy density  $\Omega_{\text{GW}}(f)$ . The strong  $n_T \approx 2.0$  blue tilt matches the nanohertz PTA anomalies and creates a definitive power-law peak at micro-Hertz frequencies inside the target sensitivity windows of LISA and DECIGO.

collaborations. The finite transition width  $\Delta a = 4 \times 10^{-4}$  limits the duration of amplification, producing a flat low-frequency floor, a sharp  $n_T \approx 2$  rise across millihertz bands, and a smooth high-frequency cutoff.

This spectral signature is uniquely accessible to LISA and DECIGO, providing a decisive observational test of rupture-driven cosmology.

#### V. CONCLUSION

We have shown that a transient hyper-stiff phase with  $w \sim 80$ , arising naturally in rupture-driven brane cosmology, generates a strongly blue-tilted primordial gravitational wave spectrum. This mechanism explains PTA anomalies and predicts a micro-Hertz peak testable by upcoming space missions, linking quantum informational space emergence with observational gravitational wave astronomy.

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